Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: Alsileli Project

Variant: Hybrid Model

No 3D scene defined, no shadings

System power: 18.00 kWp

Afiadenyigba - Ghana

Author

PVsyst V7.4.7
VC1, Simulation date:
09/04/26 10:02
with V7.4.7

Project summary

Geographical Site
Afiadenyigba
Ghana

Situation

Latitude	*6.32 °N*
Longitude	*0.87 °E*
Altitude	*66 m*
Time zone	*UTC*

Project settings

Albedo	*0.20*

Weather data
Afiadenyigba
Meteonorm 8.1, Sat=100% - Synthetic

System summary

Grid-Connected System
Simulation for year no 10

No 3D scene defined, no shadings

PV Field Orientation

Fixed plane	
Tilt/Azimuth	*30 / 90.3 °*

Near Shadings
No Shadings

User's needs
Daily household consumers
Constant over the year

Average	*1680 kWh/Day*

System information

PV Array

Nb. of modules	*30 units*
Pnom total	*18.00 kWp*

Inverters

Nb. of units	*1 unit*
Pnom total	*15.00 kWac*
Pnom ratio	*1.200*

Battery pack
Storage strategy: Self-consumption

Nb. of units	*1 unit*
Voltage	*48 V*
Capacity	*1120 Ah*

Results summary

Produced Energy	*24161 kWh/year*	*Specific production*	*1342 kWh/kWp/year*	*Perf. Ratio PR*	*75.21 %*
Used Energy	*613209 kWh/year*			*Solar Fraction SF*	*3.94 %*

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		Sheds configuration	Models used	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	30 / 90.3 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

Horizon	Near Shadings	User's needs
Free Horizon	No Shadings	Daily household consumers
		Constant over the year
		Average 1680 kWh/Day

Storage

Kind: Self-consumption
No grid reinjection

Charging strategy	Discharging strategy
When excess solar power is available	As soon as power is needed

PV Array Characteristics

PV module

Manufacturer	Jinkosolar
Model	JKM-600N-66HL4M-BDV
(Original PVsyst database)	
Unit Nom. Power	600 Wp
Number of PV modules	30 units
Nominal (STC)	18.00 kWp
Modules	3 string x 10 In series
At operating cond. (50°C)	
Pmpp	16.69 kWp
U mpp	374 V
I mpp	45 A

Inverter

Manufacturer	Huawei Technologies
Model	SUN2000-15KTL-M2-380V
(Custom parameters definition)	
Unit Nom. Power	15.0 kWac
Number of inverters	1 unit
Total power	15.0 kWac
Operating voltage	160-950 V
Max. power (=>58°C)	30.0 kWac
Pnom ratio (DC:AC)	1.20
Power sharing within this inverter	

Total PV power

Nominal (STC)	18 kWp
Total	30 modules
Module area	81.0 m²

Total inverter power

Total power	15 kWac
Max. power	30 kWac
Number of inverters	1 unit
Pnom ratio	1.20

Battery Storage

Battery

Manufacturer	Cegasa
Model	ULTRA 175 48V 1120Ah

Battery pack

Nb. of units	1 Unit
Discharging min. SOC	20.0 %
Stored energy	43.0 kWh

Battery Pack Characteristics

Voltage	48 V
Nominal Capacity	1120 Ah (C10)
Temperature	Fixed 20 °C

Battery input charger

Model	Generic
Max. charg. power	15.5 kWdc
Max./Euro effic.	97.0/95.0 %

Battery to Grid inverter

Model	Generic
Max. disch. power	24.0 kWac
Max./Euro effic.	97.0/95.0 %

Array losses

Array Soiling Losses

Average loss Fraction 4.0 %

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
5.0%	5.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	5.0%	5.0%	5.0%	5.0%

Thermal Loss factor

Module temperature according to irradiance

Uc (const) 20.0 W/m²K

Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 137 mΩ

Loss Fraction 1.5 % at STC

LID - Light Induced Degradation

Loss Fraction 2.0 %

Module Quality Loss

Loss Fraction 0.7 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Module average degradation

Year no 10

Loss factor 0.4 %/year

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

Vmp RMS dispersion 0.4 %/year

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

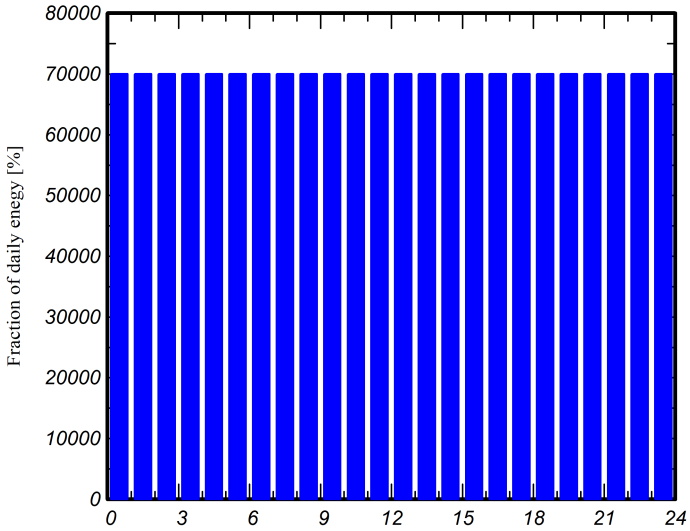
Detailed User's needs

Daily household consumers, Constant over the year, average = 1680 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Daily Energy Demand	1	70000/lamp	24.0	1680000
Stand-by consumers			24.0	24
Total daily energy				1680024

Hourly distribution



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Main results

System Production

Produced Energy	24161 kWh/year	Specific production	1342 kWh/kWp/year
Used Energy	613209 kWh/year	Perf. Ratio PR	75.21 %
		Solar Fraction SF	3.94 %

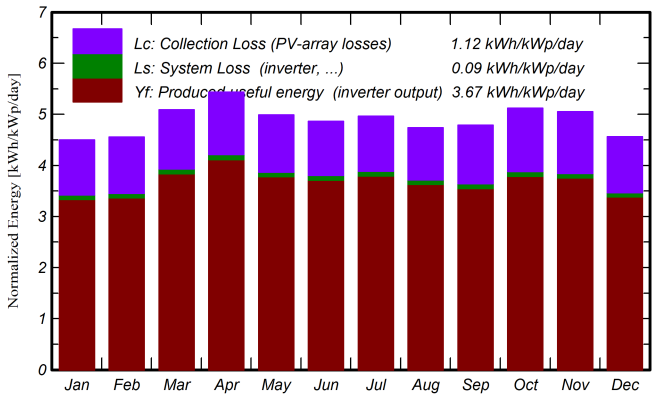
Battery aging (State of Wear)

Cycles SOW	100.0 %
Static SOW	90.0 %
Battery lifetime	10.0 years

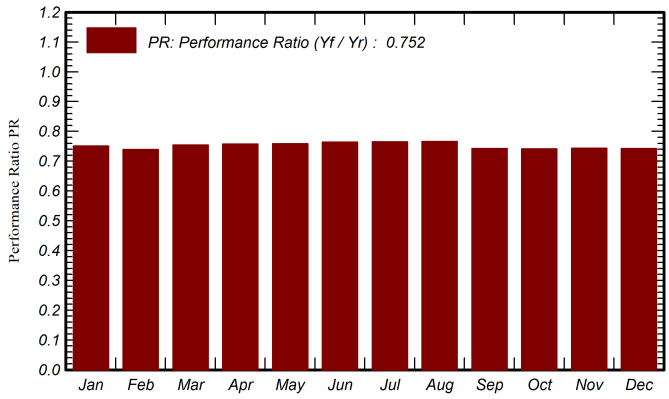
Economic evaluation

Investment		Yearly cost		LCOE	
Global	174,080.00 GHS	Annuities	0.00 GHS/yr	Energy cost	0.73 GHS/kWh
Specific	9.67 GHS/Wp	Run. costs	8,800.00 GHS/yr		
		Payback period	8.6 years		

Normalized productions (per installed kWp)



Performance Ratio PR



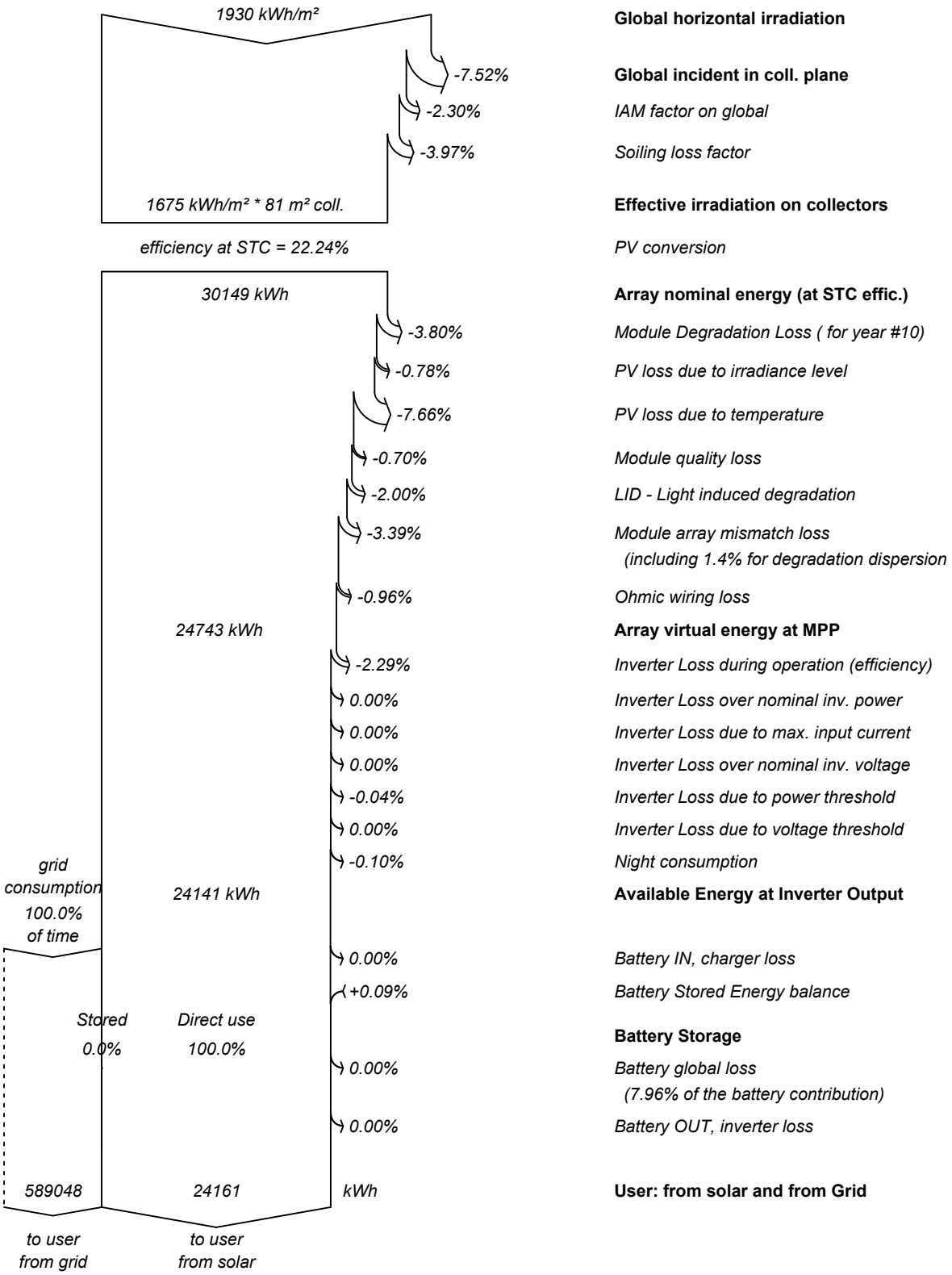
Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_User	E_Solar	EUnused	EFrGrid
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	kWh	kWh	kWh
January	152.2	87.1	27.63	139.4	129.1	1912	52081	1884	0.000	50197
February	138.7	89.4	29.00	127.7	118.4	1742	47041	1699	0.000	45342
March	169.7	100.6	29.17	157.8	149.7	2194	52081	2141	0.000	49940
April	175.2	93.9	28.54	163.1	154.9	2276	50401	2223	0.000	48178
May	166.6	86.6	28.09	154.6	146.6	2162	52081	2110	0.000	49971
June	155.4	80.9	26.35	145.9	138.3	2056	50401	2006	0.000	48395
July	168.1	84.4	26.07	153.9	145.8	2172	52081	2120	0.000	49961
August	157.5	84.2	25.45	146.9	139.3	2077	52081	2026	0.000	50055
September	157.8	67.8	25.85	143.6	133.6	1969	50401	1919	0.000	48482
October	173.4	73.2	27.01	158.7	147.5	2168	52081	2116	0.000	49965
November	162.0	79.3	27.73	151.5	140.6	2077	50401	2028	0.000	48373
December	153.3	85.4	28.22	141.4	130.8	1937	52081	1889	0.000	50192
Year	1929.8	1012.8	27.42	1784.6	1674.5	24743	613209	24161	0.000	589048

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	EUnused	Unused energy (battery full, no grid injection)
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

Loss diagram



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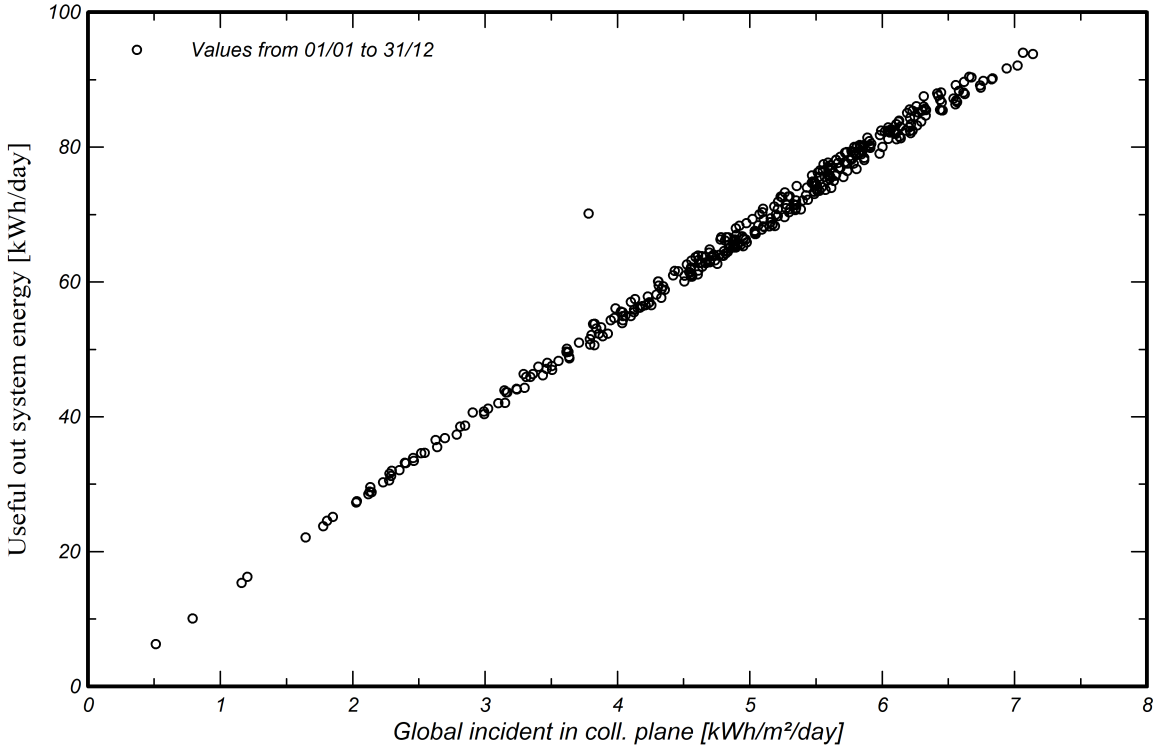
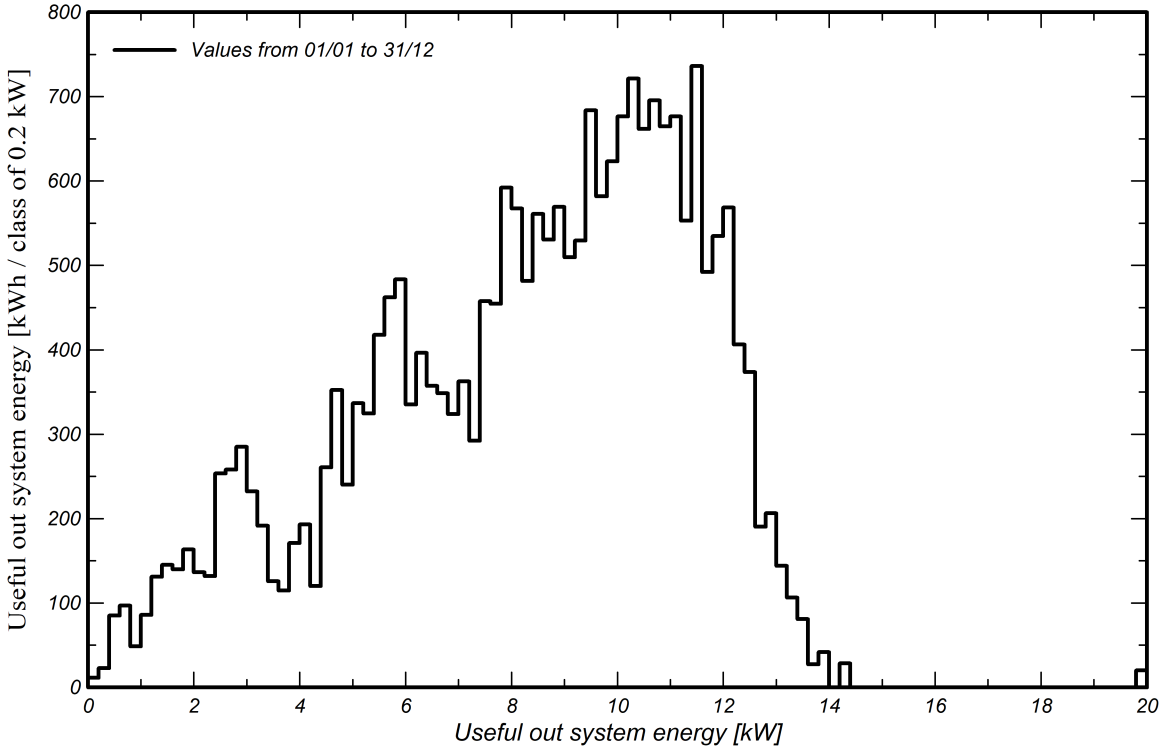
Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 8.1, Sat=100%
Kind	Monthly averages
Synthetic - Multi-year average	
Year-to-year variability(Variance)	4.7 %
Specified Deviation	
Climate change	0.0 %

Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

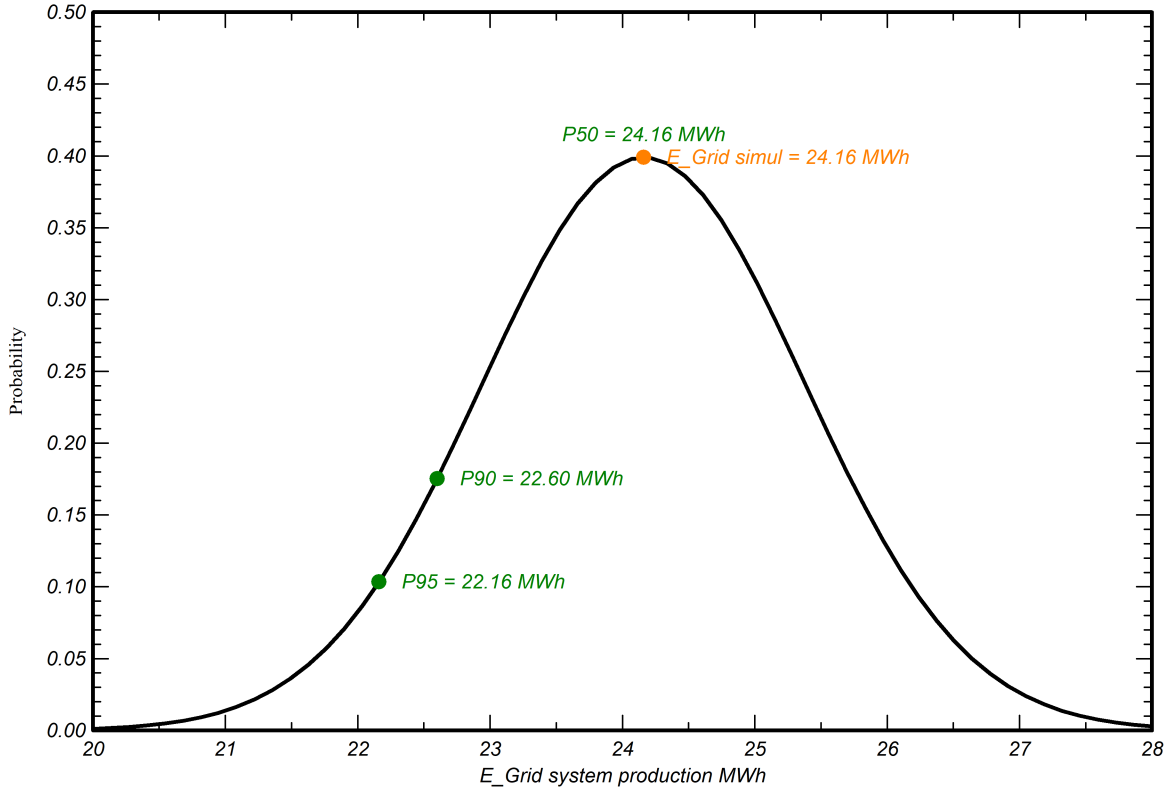
Global variability (weather data + system)

Variability (Quadratic sum)	5.0 %

Annual production probability

Variability	1.22 MWh
P50	24.16 MWh
P90	22.60 MWh
P95	22.16 MWh

Probability distribution



Single-line diagram

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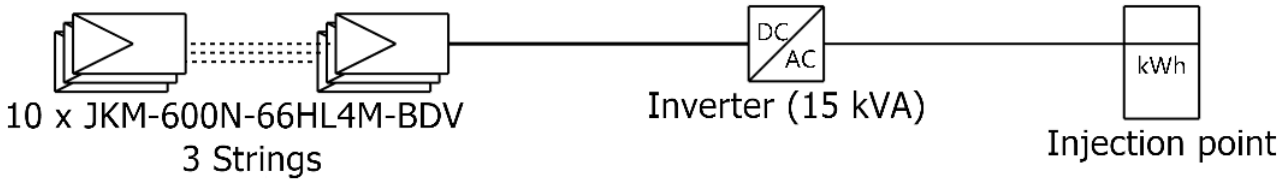

PV module	JKM-600N-66HL4M-BDV
Inverter	SUN2000-15KTL-M2-380V
String	10 x JKM-600N-66HL4M-BDV

	Alsileli Project	
	VC1 : Hybrid Model	09/04/26

Cost of the system

Installation costs

Item	Quantity units	Cost GHS	Total GHS
PV modules			
JKM-600N-66HL4M-BDV	30	1,700.00	51,000.00
Supports for modules	20	300.00	6,000.00
Inverters			
SUN2000-15KTL-M2-380V	1	14,000.00	14,000.00
Batteries			90,000.00
Other components			
Accessories, fasteners	20	25.00	500.00
Wiring	1	2,500.00	2,500.00
Combiner box	1	80.00	80.00
Studies and analysis			
Engineering	1	500.00	500.00
Installation			
Global installation cost per module	30	200.00	6,000.00
Global installation cost per inverter	1	500.00	500.00
Global installation cost per battery	1	1,000.00	1,000.00
Transport	1	2,000.00	2,000.00
		Total	174,080.00
		Depreciable asset	161,500.00

Operating costs

Item	Total GHS/year
Maintenance	
Provision for inverter replacement	2,800.00
Repairs	8,000.00
Cleaning	3,000.00
Subsidies	-5,000.00
Total (OPEX)	8,800.00

System summary

Total installation cost	174,080.00 GHS
Operating costs	8,800.00 GHS/year
Produced Energy	24.2 MWh/year
Cost of produced energy (LCOE)	0.7264 GHS/kWh

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Financial analysis

Simulation period

Project lifetime	*20 years*	*Start year*	*2027*

Income variation over time

Inflation	*0.00 %/year*
Production variation (aging)	*0.00 %/year*
Discount rate	*0.00 %/year*

Income dependent expenses

Income tax rate	*0.00 %/year*
Other income tax	*0.00 %/year*
Dividends	*0.00 %/year*

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (GHS)	Depreciable (GHS)
PV modules				
JKM-600N-66HL4M-BDV	*Straight-line*	*20*	*0.00*	*51,000.00*
Supports for modules	*Straight-line*	*20*	*0.00*	*6,000.00*
Inverters				
SUN2000-15KTL-M2-380V	*Straight-line*	*20*	*0.00*	*14,000.00*
Batteries	*Straight-line*	*20*	*0.00*	*90,000.00*
Accessories, fasteners	*Straight-line*	*20*	*0.00*	*500.00*
		Total	*0.00*	*161,500.00*

Financing

Own funds	*175,000.00 GHS*

Electricity sale

Feed-in tariff	*0.00000 GHS/kWh*
Duration of tariff warranty	*20 years*
Annual connection tax	*0.00 GHS/kWh*
Annual tariff variation	*0.0 %/year*
Feed-in tariff decrease after warranty	*0.00 %*

Self-consumption

Consumption tariff	*1.20000 GHS/kWh*
Tariff evolution	*0.0 %/year*

Return on investment

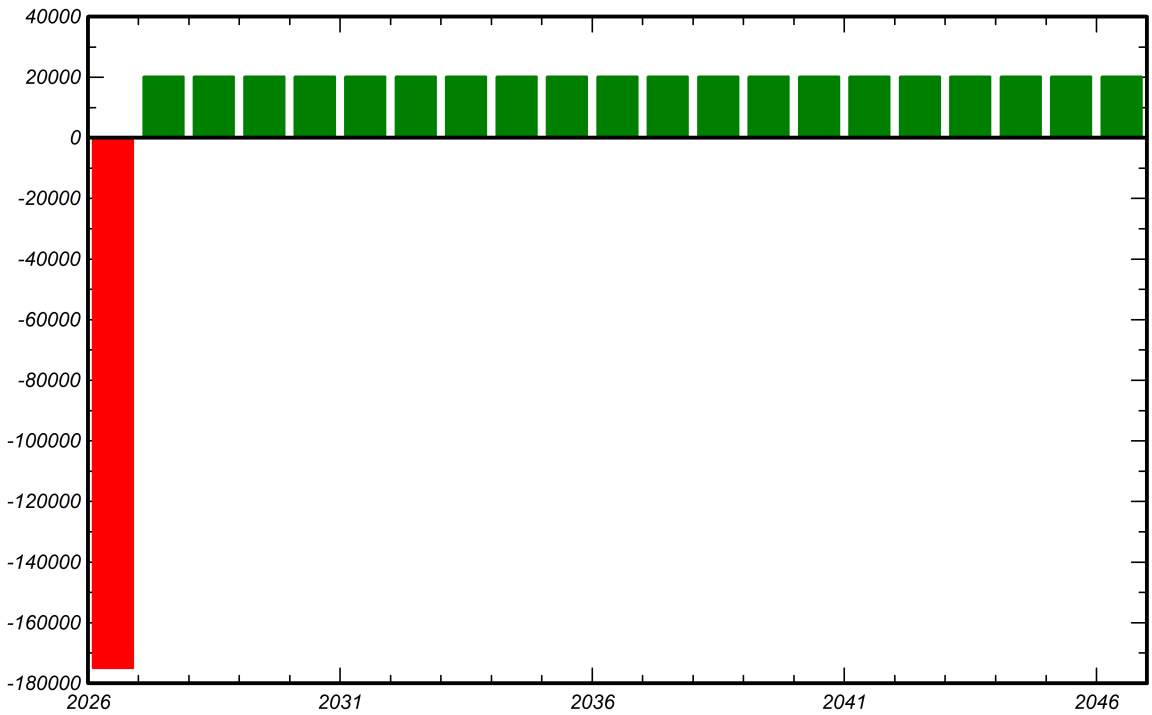
Payback period	*8.6 years*
Net present value (NPV)	*229,453.75 GHS*
Internal rate of return (IRR)	*9.77 %*
Return on investment (ROI)	*131.8 %*

Financial analysis

Detailed economic results (GHS)

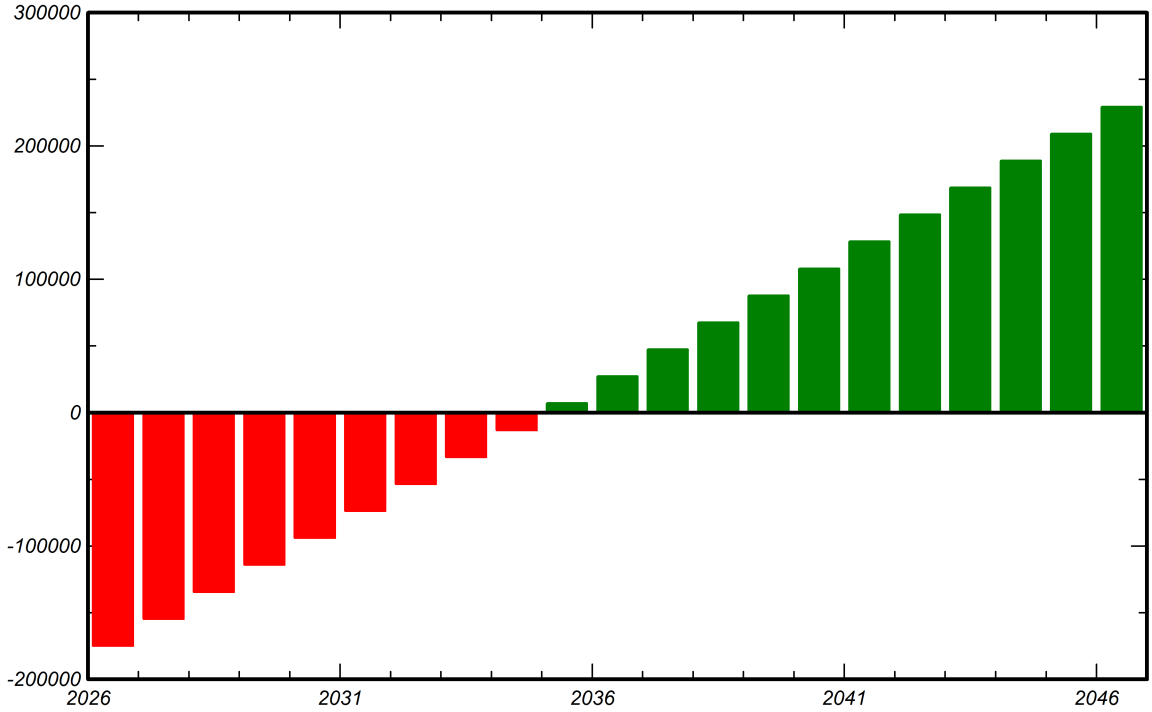
Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Self-cons. saving	Cumul. profit	% amorti.
0	0	175,000	0	0	0	0	0	0	-175,000	0.0%
1	0	0	8,800	8,075	0	0	-8,800	29,023	-154,777	11.6%
2	0	0	8,800	8,075	0	0	-8,800	29,023	-134,555	23.2%
3	0	0	8,800	8,075	0	0	-8,800	29,023	-114,332	34.9%
4	0	0	8,800	8,075	0	0	-8,800	29,023	-94,109	46.5%
5	0	0	8,800	8,075	0	0	-8,800	29,023	-73,887	58.1%
6	0	0	8,800	8,075	0	0	-8,800	29,023	-53,664	69.7%
7	0	0	8,800	8,075	0	0	-8,800	29,023	-33,441	81.3%
8	0	0	8,800	8,075	0	0	-8,800	29,023	-13,219	92.9%
9	0	0	8,800	8,075	0	0	-8,800	29,023	7,004	104.6%
10	0	0	8,800	8,075	0	0	-8,800	29,023	27,227	116.2%
11	0	0	8,800	8,075	0	0	-8,800	29,023	47,450	127.8%
12	0	0	8,800	8,075	0	0	-8,800	29,023	67,672	139.4%
13	0	0	8,800	8,075	0	0	-8,800	29,023	87,895	151.0%
14	0	0	8,800	8,075	0	0	-8,800	29,023	108,118	162.6%
15	0	0	8,800	8,075	0	0	-8,800	29,023	128,340	174.3%
16	0	0	8,800	8,075	0	0	-8,800	29,023	148,563	185.9%
17	0	0	8,800	8,075	0	0	-8,800	29,023	168,786	197.5%
18	0	0	8,800	8,075	0	0	-8,800	29,023	189,008	209.1%
19	0	0	8,800	8,075	0	0	-8,800	29,023	209,231	220.7%
20	0	0	8,800	8,075	0	0	-8,800	29,023	229,454	232.3%
Total	**0**	**175,000**	**176,000**	**161,500**	**0**	**0**	**-176,000**	**580,454**	**229,454**	**232.3%**

Yearly net profit (GHS)



Financial analysis

Cumulative cashflow (GHS)



CO_2 Emission Balance

Total: 107.5 tCO_2

Generated emissions

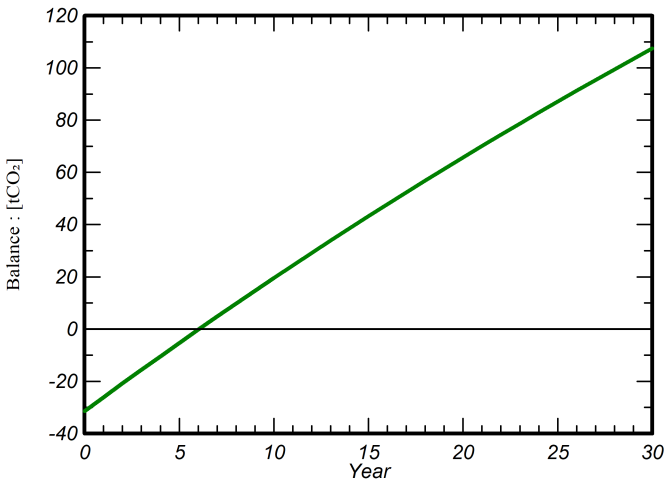
Total: 31.42 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total:	160.1 tCO_2
System production:	24.14 MWh/yr
Grid Lifecycle Emissions:	221 gCO_2/kWh
Source:	IEA List
Country:	Ghana
Lifetime:	30 years
Annual degradation:	1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	18.0 kWp	30829
Supports	1.47 kgCO2/kg	300 kg	442
Inverters	146 kgCO2/units	1.00 units	146